

AASHRITHA

Data Engineer/Data Analyst / Python Developer

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PROFESSIONAL SUMMARY

- Dynamic and motivated IT professional with 5+ years of full-time experience in Big Data Engineering with expertise in designing data-intensive applications using Hadoop Ecosystem, Big Data Analytics, Cloud Data Engineering, and Data Warehousing.
- Proficiency in writing complex SQL queries for data extraction, transformation, and analysis.
- Expertise in data Extraction, Transforming and Loading (ETL) using various tools such as SQL Server Integration Services SSIS, SSAS, SSRS.
- Developed and maintained Python-based (Pandas, NumPy) ETL processes to ensure efficient extraction, transformation, and loading of huge datasets into data warehouses.
- Expertise in developing web applications and RESTful APIs using Flask and Django frameworks, ensuring seamless integration with data processing pipelines.
- Efficiently managed and optimized data collection and processing for massive data volumes within AWS, Azure cloud services and Databricks, Snowflake frameworks.
- Expertise in cloud-based data warehousing solutions on Azure and AWS, enhancing scalability and performance.
- Hands on experience in Microsoft Azure Cloud Services, Azure Synapse Analytics, Azure SQL, Azure Cosmos DB, Kusto, Azure Blob, Data Factory, Data Bricks, Application Insights, Azure Monitoring, Key Vault, Azure Data Lake, Delta Lake, Logic Apps, Event Hub, and Function App.
- Proficient in AWS Services including IAM, EC2, EBS, VPC, S3, RDS, Redshift, Lambda, Glue, Athena, Quick Sight, Cloud Watch, Cloud Trail, Cloud Formation, SQS, SNS, and Route53.
- Hands on experience on Unified Data Analytics with Databricks, Databricks Workspace User Interface, Managing Databricks Notebooks, Delta Lake with Python and Spark SQL.
- Experience in Developing Spark applications using Spark - SQL in Databricks for data extraction, transformation, and aggregation from multiple file formats for analyzing & transforming the data to uncover insights into the customer usage patterns.
- Hands on experience in developing SPARK applications using Spark tools like RDD transformations, Spark core, Spark MLlib, Spark Streaming and Spark SQL.
- Developed ELT/ETL pipelines to move data to and from Snowflake data store using combination of Python and Snowflake Snow SQL.
- Experience in creating cloud-based data warehouse solutions with Snowflake on Azure and optimizing tables, views, and schemas for effective data retrieval and storage.
- Hands on experience in setting up workflow using Apache Airflow and Oozie workflow engine for managing and scheduling Hadoop jobs.
- Exposed the metadata to the end users using Splunk Analytics for Hadoop.
- Proficient in Hadoop ecosystem components such as HDFS, MapReduce, Hive, Pig, HBase, and Spark.
- Generated custom reports and ad-hoc analyses in MS Excel to meet specific business requirements, providing actionable insights for stakeholders.
- Strong background in building and maintaining data warehouses, creating reports & dashboards using Tableau and Microsoft Power BI, identifying the KPI's & communicating the results.
- Good understanding of NOSQL databases like Mongo DB, HBase and Cassandra.
- Expertise with implementing RBAC for user access control and an in-depth knowledge of data security and authentication techniques.
- Experience implementation of a CI/CD pipeline using Jenkins, Azure DevOps (Jira, Product Backlog) in both cloud and on-premises with GIT integration.
- Proficient with container systems like Docker and container orchestration like EC2 Container Service, Kubernetes and worked with Terraform as well.
- Well versed in Agile with SCRUM and Test-driven development methodologies.

CERTIFICATIONS:

- Microsoft Certified: Azure Data Engineer Associate (DP-203)
<https://learn.microsoft.com/api/credentials/share/en-us/AashrithaVenepally-4211/AC480072870AE2EC?sharingId=ABC52CB732637DC>
- Databricks Certified Data Engineer Associate
<https://credentials.databricks.com/429df43d-4b84-43d7-aade-a5dc844e9b1a>
- Microsoft Certified: Azure Data: Fundamentals (DP-900)
https://www.credly.com/badges/ceaf7bea-7672-45c4-8a36-f6c371b3bea2/linked_in_profile
- Microsoft Certified: Azure Fundamentals (AZ-900)
https://www.credly.com/badges/3dddec67-9414-4e17-b6c1-1b7b1fb0d107?source=linked_in_profile
- JIRA Fundamentals Badge
<https://university.atlassian.com/student/award/pQmXqnx2FCUKfNznFxmWd0n>

TECHNICAL SKILLS:

- Programming Languages: SQL, Python, Java, Scala
- Big Data Tools: Hadoop, Map Reduce, Spark, Flink, Nifi, HBase, Hive, Pig, Sqoop, Kafka, Oozie, Autosys
- Databases: Oracle, MY SQL, SQL Server, NO SQL-MongoDB, Dynamo DB, Cassandra, HBase
- Data Warehousing: Microsoft Azure Synapse Analytics, Amazon Redshift, Google Big Query, Snowflake
- ETL Data Processing Frameworks: Apache Hadoop, Apache Spark, Apache Airflow, Databricks, Snap logic.
- Cloud Platform: Azure, AWS, GCP
- BI Tools: SSIS, SSRS, SSAS, MS Excel, Microsoft Power BI, Tableau, MS Visio
- Version Control: Git
- CI/CD Tools: Azure DevOps, Jenkins, AWS Code Pipeline

PROFESSIONAL EXPERIENCE:

Barclays, NJ

Feb 2023 – Till Date

Data Engineer

Responsibilities:

- Proven experience as an Azure Data Engineer, with a focus on Azure Synapse and Azure Data Factory.
- Implemented data validation scripts in Python to identify and rectify inconsistencies in incoming data, improving overall data quality.
- Utilized Python libraries (e.g., Pandas, NumPy) for data manipulation, analysis, and transformation tasks.
- Proficient in designing and implementing scalable data pipelines using Azure Data Factory for efficient data transfer.
- Created Pipelines in Azure Data Factory (ADF) using Linked Services/Datasets/Pipeline/ to Extract, Transform, and load data from different sources like Azure SQL, Blob storage, Azure SQL Data warehouse, using Azure Key Vaults to store secrets.
- Skilled in utilizing Azure Databricks to perform advanced data processing and transformation tasks, ensuring data readiness for downstream systems.
- Designed and automated data workflows using Azure Logic Apps, event hub in streamlining data ingestion and integration across various Azure services and external systems.
- Strong proficiency in Kusto Query Language (KQL) for querying, data extraction and transformation.
- Demonstrated proficiency in using Azure Synapse Analytics for advanced analytics and data warehousing, combining structured and unstructured data for actionable insights.
- Developed scalable data pipelines using Azure Databricks, enabling efficient processing and transformation of massive data volumes.
- Streamlined data processing, cleansing, and enrichment with Stream Sets, enhancing data quality.
- Orchestrated data workflows, automated data movements, and transformations using Azure Logic Apps, Functions, and Stream Sets, optimizing data integration processes.

- Enabled real-time data ingestion from Apache Kafka and Azure Event Hubs using Stream Sets, facilitating timely insights and updates.
- Contributed significantly to data governance initiatives and access controls, ensuring data privacy and security within the Azure ecosystem (IAM, Active Directory, Purview).
- Demonstrated expertise in leveraging Snap Logic's cloud-native architecture for building and deploying ETL processes in cloud environments, including AWS, Azure, and Google Cloud.
- Implemented data lineage and metadata management solutions, providing comprehensive visibility into data movement and transformations.
- Utilized Jira for comprehensive issue tracking, monitoring, and reporting, resulting in improved issue resolution times.
- created captivating data stories in Tableau and Power BI by integrating annotations and graphics to convey important insights to audiences irrespective of their technical knowledge.

Environment: Python, PySpark, SQL, KQL, Databricks, Azure Data Factory, Synapse Analytics, Logic Apps, Event Hub, Function App, IAM, Azure AD, Key vault, Azure DevOps, Azure Purview, Spark, Hive, Apache Kafka, Snap Logic, Power BI, Tableau, GIT, Agile.

Black Rock, Wilmington, DE

Sep 2022 – Dec 2022

Data Analytics Engineer Intern

- Implemented end-to-end data pipelines using Azure Data Factory to extract, transform, and load (ETL) data from diverse sources.
- Designed and implemented data processing workflows using Azure Databricks, leveraging Spark for large-scale data transformations.
- Developing and maintaining data pipelines to extract, transform, and load (ETL) data from various sources into Azure Synapse.
- Built scalable and optimized Snowflake schemas, tables, and views to support complex analytics queries and reporting requirements.
- Leveraged Azure Data Lake Storage as a data lake for storing raw and processed data, implementing data partitioning and data retention strategies.
- Utilized Azure Blob Storage for efficient storage and retrieval of data files, implementing compression and encryption techniques to optimize storage costs and data security.
- Integrated Azure Data Factory with Azure Logic Apps for orchestrating complex data workflows and triggering actions based on specific events.
- Implemented data replication and synchronization strategies between Snowflake and other data platforms using Azure Data Factory and change data capture techniques.
- Developed and deployed Azure Functions for data preprocessing, data enrichment, and data validation tasks in data pipelines.
- Implemented advanced analytics and machine learning workflows using Azure Machine Learning and Snowflake, enabling predictive analytics and data-driven insights.
- Designed and implemented data archiving and data retention strategies using Azure Blob Storage and Snowflake's Time Travel feature.
- Developed custom monitoring and alerting solutions using Azure Monitor and Snowflake Query Performance Monitoring (QPM) for proactive identification and resolution of performance issues.
- Integrated Snowflake with Power BI and Azure Analysis Services for creating interactive dashboards and reports, enabling self-service analytics for business users.
- Implemented data cataloging and data lineage solutions using tools like Azure Purview to provide a comprehensive understanding of data assets and their relationships.
- Collaborated with cross-functional teams including data scientists, data analysts, and business stakeholders to understand data requirements and deliver scalable and reliable data solutions.

Environment: ETL, Azure Databricks, Data Factory, Logic Apps, Snowflake, Functional App, Blob Storage, Snowflake, MS SQL, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, GIT, JIRA, Jenkins, Kafka, ADF Pipeline, Power BI, Agile.

Cigna, India

Apr 2020 -Dec 2021

Hadoop Developer

Responsibilities:

- Utilized agile software development framework SCRUM as a lightweight process to manage and control software product development.
 - Skilled in cluster analysis using different big data analytic tools like Hive, and MapReduce.
 - Used open-source Python web scraping framework to crawl and extract data from web pages.
 - Proficient in connecting Flask and Django applications to databases, optimizing data retrieval, and implementing efficient data storage solutions.
 - Used Sqoop to import data from a mainframe dataset to HDFS.
 - We also showed how to import data from various data sources (Oracle, DB2) into Hadoop and used Hive, MapReduce to perform the conversion.
 - Experience in migrating data using Sqoop from HDFS to relational database system and vice-versa.
 - Experienced in configuring Flume to stream data into HDFS.
 - Extracted the daily sales, hourly sales, and product mix of items sold in the store and loaded them into the global data warehouse.
 - Have populated many XML files into Hadoop using MapReduce's DOM parser.
 - Created Airflow Scheduling scripts in Python. Expert in Zookeeper, Oozie, and Data pipeline operational services to coordinate clusters and plan workflows.
 - Responsible for creating Hive tables, loading structured data from MapReduce jobs into tables, and creating Hive queries to further analyze logs to identify problems and behavior patterns.
 - Used python sub-process module to perform UNIX shell commands and extracted data from Agent Nodes into HDFS using Python scripts.
 - Developed scripts and batch jobs for scheduling various Hadoop programs and was involved in the maintenance and review of Hadoop log files.
 - Involved in data import and export from RDBMS to HDFS and vice versa using Sqoop.
 - Extracted the required data from server into HDFS and Bulk Loaded cleaned data into HBase.
- Environment: Oracle, HDFS, Map Reduce, Hive, HBase, Python, Django, Unix, RDBMS, Zookeeper, Oozie, MapReduce, Airflow.

Deutsche Bank, India

Dec 2018 - Mar 2020

Assistant Systems Engineer (Data Analyst)

Responsibilities:

- Worked as a Python Developer/ Data Analyst in Big Data - Analytics & Insights project (Agile) where we store, process & manage huge amounts of bank transaction data to generate insightful visual reports.
- Developed python code/enhancement according to business requirements and tested against historic data via source systems.
- Developed and maintained ETL scripts using Python and SQL to transform data into meaningful insights.
- Prepared automation scripts using Hive interface and python modules.
- Worked with Stream Sets tool & implemented complete big data pipelines for batch and real time processing using PySpark (ETL Pipelines).
- Worked on multiple users sending in various types of data files, analyzing, testing, and validating data and exploring datasets using NumPy, Pandas, Matplotlib, Requests, Scikit-learn libraries.
- Implemented data security measures using AWS IAM policies and S3 bucket policies to ensure compliance with regulatory requirements.
- Created and maintained data warehouses using AWS Redshift, AWS Glue for data crawling, Athena for querying and Quick Sight to support business intelligence reporting and analytics.
- Involved in Data Visualization and identifying Key Performance Indicators (KPI's) bench marking using Tableau.
- Performed SDLC tasks using GIT, JIRA, Jenkins.

- Prepared test evidence documentation and artifacts.
 - Created complex DAX formulas for calculated columns and measures, supporting advanced analytics within Power BI.
 - Discussions with Stakeholders on gathering requirements, Analysis, data sources on boarding & data validation to drive out with a great value of data.
 - Crafted compelling data stories in Power BI and Tableau by combining visualizations and annotations to communicate key findings to both technical and non-technical audiences.
 - Implemented Jenkins pipelines to automate the build, test, and deployment processes, reducing manual errors and increasing deployment frequency.
- Environment: Python, SQL Server, MySQL, T-SQL, Oracle DB, SSIS, SSRS, MS Excel, DAX, AWS, IAM, EC2, S3, Redshift, Lambda, Glue, Athena, Tableau, Power BI, Agile, Git, Jira, Jenkins

HSBC, India

Jan 2018– Nov 2018

SQL Developer

Responsibilities:

- Expertise in Dimensional Data Modeling for Data Mart design, identifying Facts and Dimensions, and developing fact tables, dimension tables using Slowly Changing Dimensions (SCD).
- Designed and implemented complex ETL processes using SSIS for efficient data extraction, transformation, and loading.
- Built financial models in MS Excel to support budgeting, forecasting, and decision-making processes, incorporating complex formulas and data validation techniques.
- Expertise in designing ETL data flows using SSIS, creating mappings/workflows to extract data from SQL Server and data migration and transformation from Access/Excel Sheets using SQL Server SSIS.
- Developed and maintained SSRS reports for diverse business requirements, enhancing data visualization and reporting capabilities.
- Created and maintained Shell scripts for routine database maintenance tasks, including backup and restore operations.
- Collaborated with business analysts to gather requirements and translate them into SQL queries and stored procedures.
- Utilized advanced T-SQL skills for data manipulation, stored procedures, triggers, and performance optimization.
- Implemented data security measures and managed user access, ensuring compliance with company policies.
- Created interactive dashboards and reports using Tableau and Power BI for key stakeholders.
- Conducted performance tuning on existing SQL queries, for improving the database efficiency.
- Designed and maintained data models to support reporting needs and ensure data accuracy.
- Collaborated with cross-functional teams to identify and implement business intelligence solutions.

Environment: Python, ETL, SQL Server, MySQL, T-SQL, SSIS, SSRS, Tableau, Power BI, MS Excel, MS Office, SCD, Shell scripting, Agile, Git, Jira.

Education:

Master's: M.S in Information Systems Technology at Wilmington University -2022 with 4 GPA

Bachelor's: BTech in Computer science and Engineering at JNTUH – 2018 with 3.7 GPA